

WEEK 1: INTEGRATIVE PHYSIOLOGY

IMMUNITY, INFECTION AND INFLAMMATION

IMMUNITY

OVERVIEW

- Composed of **specialized effector cells** (detect and respond to foreign antigens) and **regulatory molecules** (wide range of soluble mediators or cytokines)

- Immune System

- Clearing senescent or abnormal self-cells
- Sometimes misdirected attacks on normal tissues (autoimmune diseases)

- Innate immunity

- Internal defense mechanism
- Receptors recognize sequences of sugars, lipids, amino acids, or nucleic acids
- Once the system detect invaders, it activates defense mechanisms such as:
 - Release of cytokines (interferons) - regulatory molecules (signals)
 - Phagocytosis
 - Production of antibacterial peptides
 - Activation of the complement system (proteins that help destroy pathogens)
 - Several proteolytic cascades (series of reactions)
- First line of defense in humans
 - *Antigen* - any foreign material
 - *Antibody* - binds with antigen, so that the body recognizes it
- Effects are mediated by cytokine release, complement activation, and other systems

- Adaptive acquired immunity

- Stronger defense mechanism
- T and B lymphocytes are activated by specific antigens
 - **B cell** - form clone that create antibodies which attaches to foreign invaders and attack them
 - **T cell** - have antigen-specific receptors to detect antigen. Upon activation, it proliferate and produce cytokines
- After the infection died, **memory cells** remain
 - If the same infection shows up again, these memory cell will remember it and helps the body respond faster and more powerfully

INNATE IMMUNITY

- Can detect wide variety of signals from harmful invaders like bacteria, viruses, tumors, and even transplanted cells

- Present in all multicellular organisms

- Examples:

- Eosinophil, epithelial barrier, mast cell, basophil, NK cell, neutrophil, dendritic cell, complement, macrophage

ADAPTIVE (ACQUIRED) IMMUNITY

- Highly specific immune response where lymphocytes makes antibodies (B cells) or cell-surface receptors (T cells)

- Two key parts:

- Humoral immunity
 - B cells produce antibodies called *immunoglobulins* when they encounter antigens, like bacterias' or viruses' protein on its surface
 - Antibodies produced by plasma cells (differentiated B cells)
 - This circulates in the blood to bind pathogen and neutralize it, and activate complement system to destroy invaders
 - Major defense against bacterial infections
- Cell-mediated immunity
 - Controlled by T cells, which focuses more on infected cells or foreign tissue
 - Types of T cells
 - Cytotoxic T cells (CT8): directly kills infected or cancerous cells by inserting perforins, which creates hole in the membrane, and then triggering apoptosis
 - Helper T cells (CT4): releases cytokines
 - Regulatory T cells: controls immune response
 - Memory T cells: remember previous infection
 - For delayed allergic reactions, transplant rejections, antiviral, antifungal, antitumor defense

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- Defense against viruses, fungi, and certain bacteria (e.g., mycobacterium tuberculosis)

LYMPHOCYTES

- Essential cells specifically for acquired immunity
- Formed in bone marrow, and matures in thymus, lymph nodes, and spleen
- Travel via lymphatic
- Lymphocyte development
 - T cells mature in thymus (10% succeed; 90% will die)
 - B cells matures in bone marrow
 - Monocyte matures in macrophage
 - Natural killer cells also forms in bone marrow and liver
 - NKT cells form in thymus, sharing traits of T and NK cells
 - T and B cells move to lymph nodes to help coordinate immune responses
- Continuously circulates between the bloodstream (2%) and lymphoid organs (98%)
- Types of lymphocytes:
 - B cells - primarily responsible for humoral activity, which produces antibodies;
 - Plasma cell - B cell will first differentiate into immunoglobulin-producing cells, and would later become this
 - T cells
 - Cytotoxic T cells - killer cells that inserts perforins (only found inside the cells), aided by helper T cells
 - Helper T cells - assist other immune cells by producing cytokines
 - Th1 cells (IL-2, γ -interferon) - stimulates cellular immunity
 - Th2 cells (IL-4, IL-5) - help activate B cells to produce antibodies
 - Th17 cells (IL-6, IL-17) - recruits neutrophils; involved in autoimmune inflammation
 - Treg cells (IL-10) - suppresses T cell-driven responses
 - Suppressor T cell - guard t-cells to not attack organs
 - Memory T cell

- NK cells - part of innate immune system, which kills infected or abnormal cells without the need of recognizing specific antigens
- NKT cells - hybrid between T and NK cells

MEMORY B CELLS AND T CELLS

- Formed after the immune system exposed to an antigen
- "Remember" the antigen and persists in the body after the initial infection is cleared
- Memory response can last a long time, and sometimes, life-long
- Converts rapidly into effector cells upon re-exposure to the same antigen
- Movement and location of lymphocytes:
 - After the B and T cells activates in the lymph nodes, it is dispersed through the body where they are mostly needed (e.g., mucosal tissue or gastrointestinal tract)
 - Ensures that memory cells are located close to sites of potential reinfection
 - Chemokines (chemotactic cytokines) helps direct memory cells to specific sites

ANTIGEN RECOGNITION

- Develops even before the body ever encounters a specific antigen
- Antigen enters the body, and binds directly to appropriate receptor on B cells
- **B cell:** antigen can bind directly, but B cells contact helper T cells to fully activate and start producing antibodies
- **T cell:** antigen is first taken up by **antigen-presenting cell (APC)** which where it will be digested, and a peptide fragment is displayed to cell receptors
- **Clonal selection** - This activates T cells, and starts to divide and clone many copies of B cell to target the same antigen
 - **Negative selection** - any lymphocyte that are active to its own normal, healthy cells are deleted
 - Immune tolerance - preventing lymphocytes from attacking the body's own tissues
 - *If failed*, lupus, multiple sclerosis, rheumatoid arthritis happens

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ANTIGEN PRESENTATION

- APCs are important to present antigens to T cells to start immune response

- Include dendritic cells (lymph nodes & spleen), Langerhans cells (skin), macrophages, B cells, and epithelial cells (e.g., intestinal lining)

- APCs present antigen peptide fragments bound to HLA proteins encoded by MHC genes

- Major Histocompatibility Complex (MHC) - special protein used by APCs to show pieces

- Class I MHC
 - Found in nucleated cells; present peptides from proteins made inside the cell
 - Processed inside and displayed on the cell surface using MHC Class I molecules
 - Basically, when the inside of the cell is infected, class I MHC presents fragment of the infected pieces to be able to activate CD8 (Cytotoxic T cell) to attack those infected pieces
- Class II MHC
 - Found mainly on professional APCs (B cells, dendritic cells, activated T cells)
 - Present peptides from extracellular antigens that were engulfed by the cell (via endocytosis)
 - Broken down in endosomes and displayed on the cell surface using MHC Class II molecules
 - Basically, class II MHC presents the pieces that the immune system has already attacked before. CD4 (helper T cells) recognize this to activate immune response

T CELL RECEPTORS TYPES

- CD8 (found in cytotoxic T cells) - recognize MHC-I proteins

- CD4 (found in helper T cells) - recognize MHC-II proteins

- Helps in the binding of MHC proteins to T cell receptors and aid lymphocyte development

- Activated CD8 T cells - directly kills targets

- Activated CD4 T cells - secrete cytokines to activate other immune cells

- Immunologic synapse - T cell receptor surrounded by adhesion molecules & proteins that connect with complementary proteins on APCs

- Two signals are required for activation: digested antigen binding to T cell receptor and interaction of surrounding proteins in the synapse
- If first signal happens without the second, T cell becomes inactive and unresponsive

B CELLS

- Directly bind to antigens, but needs help from helper T cell (Th2 subtype) to become fully activated and produce antibodies

- Cytokine influence

- IL-4 - promotes Th2 lineage (supports B cell help)
- IL-12 - promotes Th1 phenotype
- IL-2 - acts autocrinely to stimulate T cell proliferation

- Once it divides, it differentiate into memory B cells (long-term immunity) and plasma cell (secrete large amount of antibodies)

IMMUNOGLOBULINS

- Bind and neutralize protein toxins, block virus/bacteria attachment to host cells, opsonizes bacteria (enhance phagocytosis), and help activate complement system

- Five main types:

Immunoglobulin	Function	Heavy Chain	Additional Chain	Structure	Plasma Concentration (mg/dL)
IgG	Complement activation	$\gamma_1, \gamma_2, \gamma_3, \gamma_4$		Monomer	1000
IgA	Localized protection in external secretions (tears, intestinal secretions, etc.)	α_1, α_2	J, SC	Monomer; dimer with J or SC chain; trimer with J chain	200
IgM	Complement activation	μ	J	Pentamer with J chain	120
IgD	Antigen recognition by B cells	δ		Monomer	3
IgE	Reagin activity; releases histamine from basophils and mast cells	ϵ		Monomer	0.05

- Composed of 4 polypeptide chains: 2 long chain (heavy chains) and 2 short chains (light chains) - connected by disulfide bridges for flexibility

- Contains Variable (V), Diversity (D), Joining (J), Constant (C) segments:
 - Fab (arms) portion - contains variable region that binds to antigen (V, D, J)
 - Fc (stem) portion - contains effector function (C)

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SOLUBLE REGULATORS OF IMMUNITY

COMPLEMENT SYSTEM

- Consisting of over 30 plasma proteins that work together to kill pathogens

- "Complements" the action of antibodies

- Pathways:

- Classical Pathway
 - Triggered by immune complexes (antigen-antibody)
 - Activated by adaptive immune system
- Mannose-Binding Lectin Pathway
 - Triggered when lectin binds bacterial mannose groups
 - Activated by innate immune system
- Alternative (Properdin) Pathway
 - Triggered by direct contact with various pathogens
 - Part of innate immune system, and can also activate without antibodies

- Functions:

- Opsonization
 - Coats pathogens with complement proteins which makes the pathogen easier to engulf
- Chemotaxis
 - Attract immune cells to the site of infection
- Cell lysis
 - Create holes in the membrane through insertion of perforins for ion imbalance to disrupt membrane
- Bridges innate and acquired immunity by activating B cells & supporting immune memory
- Clears waste after apoptosis

OTHER IMMUNE/INFLAMMATORY EFFECTOR CELLS

GRANULOCYTES

- Contain cytoplasmic granules with substances that help with inflammation and allergic reactions

- Types:

- Neutrophils
 - Produced in the bone marrow and live for about 6 hours in the bloodstream

- Body needs to produce over 100 billion per day to keep the right amount of neutrophils
- Can cause host tissue damage in diseases (e.g., rheumatoid arthritis)

- Eosinophils

- Usually found in GI, respiratory, and urinary tracts
- **Specialize in parasitic infection**
- Involved in allergic reactions like asthma, and some GI/respiratory diseases

- Basophils

- Enter tissues, release proteins & cytokines
- Similar to mast cells - activated when allergen triggers, then releases histamine when activated
- Activated by antigens binding to IgE on cell surface
- Involved in allergic reaction (e.g., rashes, hay fever, or anaphylactic shock)
- Allergens = harmless substances for most people but trigger IgE production in sensitized individuals

MAST CELLS

- Heavily granulated cells found in connective tissue that usually come into contact with the outside world

- Granules inside contains:

- Proteoglycans, histamine, various proteases

- To activate, allergen bind to IgE on the cell surface

- Also involved in inflammatory responses via IgE and IgG

- Releases histamine and other chemicals
 - Acquired immunity: help combat invading parasites
 - Innate immunity: When encountering bacterial products, they release TNF- α without antibody involvement

- Degranulation can cause allergic symptoms, including anaphylaxis in severe cases

MONOCYTES

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- Type of WBC that originate in the bone marrow, and spends 72 hours circulating in the bloodstream
- Once in the tissue, turns into macrophages - large, powerful cells that can engulf
 - Lifespan isn't well known, but bone marrow transplant data says they can live for about 3 months; do not reenter circulation
 - Different macrophages in the body:
 - Kupffer cells - liver
 - Pulmonary alveolar macrophages - lungs
 - Microglia - brain
 - May become multinucleated giant cells in chronic inflammation (e.g., tuberculosis)
 - Activated by cytokines, released from T cells
 - Respond to chemotactic signals, engulf and kill bacteria (similar to neutrophils)
 - **Key players in innate immunity**
 - Secretes: factors influencing lymphocytes and other cells, prostaglandins (E series), and clot-promoting factors

PLATELETS

- Non-nucleated cells found in the blood important for hemostasis - process of stopping bleeding
- Not immune cells, but cooperate with inflammatory cells in tissue injury response
- Formed in bone marrow as large fragments called *megakaryocytes*
- Thrombopoietin - controls platelets production and is produced by the liver and kidneys, which acts to stimulate megakaryocytes to produce platelets
- Have receptors in membrane for various molecules:
 - Collagen, ADP, von Willebrand factor, fibrinogen
- von Willebrand factor
 - Produced by endothelial cells and platelets
 - Aids platelet adhesion and regulated factor VIII levels
- Activation & aggregation
 - When injury is exposed, platelets stick to the exposed collagen and von Willebrand factor (vWF) in the vessel wall
 - Platelets have receptors for vWF, which helps them stick to the injury site
 - Once it sticks to the injury site, it releases its granule contents, such as ADP

- ADP binds to P2Y1, P2Y2, P2X1 receptors to recruit more platelets
- Platelet-activating factor (PAF) - help platelets clump together
 - Promotes aggregation & inflammation
- Thromboxane A2 production - regulates clotting balance
- Thrombocytopenia (low platelets) - easy bruising
 - Thrombasthenic purpura - subcutaneous hemorrhages, normal count but abnormal platelets
 - Thrombocytosis (high platelets) - risk of thrombosis -> abnormal blood clots which cause heart attacks or strokes

IMMUNITY

LOCAL INJURY & INFLAMMATION

Basically, our body releases **cytokines** that trigger inflammatory response. **Neutrophils** then move into the affected area. **Adhesion molecules** help stick to the blood vessel walls and move out of the tissues. **Complement proteins** and **IgG antibodies** also get involved in attacking and marking harmful antigens. **PAF** plays a role in the early stages of inflammation, promoting movement of cells to the area. As inflammation continues, **monocytes**, and **lymphocytes** join the response. **Arterioles dilate**, allowing more blood flow. Then, **capillary's permeability increases** or becomes leakier, allowing immune cells and proteins to move into the tissue more easily. As inflammation arises, skin experiences **redness, swelling, tenderness, and pain**.

SYSTEMIC RESPONSE TO INJURY

- Triggered by cytokines from inflammation, injury, or infection
- Alters acute phase protein levels ($\geq 25\%$ change), mostly from liver
- Example: C-reactive protein (CRP) - activates monocytes for more cytokine release
- Other systemic effects: somnolence, negative nitrogen balance, fever

WOUND HEALING

- When tissue is damaged, platelets sticks to exposed collagen and laminin via integrins

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- Thrombin promotes platelet aggregation and granule release
 - Platelet granules create inflammatory response
- The inflammatory response signals the endothelial cells to put selectins on the surface
- WBC are attracted to the area by selectins
 - Extravasation - process by which they attach to integrins on endothelial cells and move through the blood vessel walls
- Once in the tissue, cytokines activate **fibroblasts**, **epithelial cells**, and **macrophages** necessary for wound healing and scar formation
- **Plasmin** - helps healing process by breaking down fibrin (clotting protein), allowing keratinocytes to move into the wound and restore epithelium beneath the scab

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EXCITABLE TISSUE: NERVE

THE NEURON: BASIC WORKING UNIT OF THE NERVOUS SYSTEM

- **Cell body (soma)** - contains nucleus (metabolic center and stores hereditary material or DNA)
- **Dendrites** - extend outward from the soma (receives and process signals sending it to the soma)
- **Axon** - originates from the axon hillock (thickened area of soma)
 - Initial segment - first portion of axon
- **Presynaptic terminal** - has synaptic knobs (boutons)
 - Contain vesicles of neurotransmitters
- **Myelin sheath** - protein-lipid complex that wrapped around the axon
 - **Schwann cells** - found in PNS
- **Nodes of Ranvier** - in between axon; gaps
- **Four important zones:**
 - Dendritic zone (dendrites) - where multiple local potential changes generated by synaptic connections are integrated
 - Trigger zone (axon hillock) - generates APs (initial segment)
 - Axon - transmits APs to terminals
 - Nerve endings (axon terminal) - release neurotransmitters when APs arrive
- **Unipolar** - have one process
- **Bipolar** - have two specialized processes
 - Dendrites - carries information to cell; axon - transmits information from cell
- **Pseudounipolar** - a single process splits into two, both of which functions as axons
- **Multipolar** - one axon and many dendrites

EXCITATION & CONDUCTION

- Neurons respond to electrical, chemical, or mechanical stimuli
- Responses involve change in ion conduction across membrane
- Types:
 - **Nonpropagated potentials (localized)**
 - Do not travel long distances
 - Synaptic, generator, electric potentials
 - **Propagated potentials**
 - Action potentials, the main way neurons send messages
- Impulse conduction:

- APs travel along axon to its termination
- Conduction is active, self-propagating, maintaining constant amplitude & velocity

RESTING MEMBRANE POTENTIAL

- Voltage difference between inside and outside of **neuron = -70 mV**, close to K equilibrium potential
- Must have an unequal distribution across the membrane (concentration gradient). and have membrane permeability to ions (via ion channels/pores)
- **Outside is positive (more Na) = inside is negative (more K)**
 - RMP is maintained by Na/K ATPase pump
 - Inside is naturally negative since there are negatively charged proteins trapped inside the cell
 - Membrane is more permeable to K than Na at rest
 - Leak channels contribute to the difference in ion concentration across the membrane during resting membrane potential (more K leak channels)
- Membrane is said to be polarized
- Small, but measurable, voltage exists across the membrane

ALL-OR-NONE ACTION POTENTIALS

- Threshold intensity - minimum amount of electrical current needed to trigger an action potential
 - Depends how long the stimulus last: weak stimulus = longer time for triggering action potential, and vice versa
- **Adaptation** - when the nerve gets used to stimulus increasing slowly
- **All-or-none - stimulus is weak = no action potential**
 - Weak and fixed stimulus also affects membrane potential
 - It causes a small, local, depolarization
 - If the stimulus is a positive charge (anodal current), it can also make the membrane more negative (hyperpolarization) - K efflux, Cl influx
 - This small changes are called electronic potentials

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- If stimulus keeps increasing, it will reach -55 mV, which will turn to threshold potential, and becomes action potential

CONDUCTION OF ACTION POTENTIALS

Starts at the resting membrane. When action potential hits and reaches threshold, depolarization happens (inside is positive; outside is negative), where Na channels open. Once the Na channels reach equilibrium (60 mV), it starts to become inactive, then K channels start to open. After which, Na channels become inactive, and K channels open more slowly for efflux, and will cause repolarization. Shortly after, K channels will open very fast, which will cause hyperpolarization. Na/K ATPase will then help in balancing and bringing RMP back to equilibrium.

ION-SPECIFIC INFLUENCES

- Na: lower extracellular Na will contribute to little effect on resting potential (low Na permeability at rest)
- K:
 - Hyperkalemia (higher extracellular K) - resting potential closer to threshold. Thus, neuron becomes more excited
 - Hypokalemia (lower extracellular K) - hyperpolarization. Thus, neurons become less excited
- Ca: inversely proportional
 - Higher extracellular Ca - decreased excitability (membrane stabilization)
 - Lower extracellular Ca - increased excitability (less depolarized needed)

REFRACTORY PERIOD

- Threshold goes up - when membrane is hyperpolarized (hard to trigger AP)
- Threshold goes down - when membrane is depolarized (easier to trigger AP)
 - **Absolute refractory**
 - Starts as soon as the neuron hits the threshold
 - Lasts about $\frac{1}{3}$ of the way through repolarization
 - No new AP can propagate

- **Relative refractory**

- Followed after absolute refractory
- Lasts until membrane finishes repolarizing and reaches after-depolarization
- New action potential can happen if the stimulus is stronger than usual

NERVE FIBER TYPES & FUNCTION

- **Group A** - largest and fastest
- **Group B** - smaller and slower
- **Group C** - smallest and slowest (unmyelinated)
 - Larger diameter, faster conduction
 - **Large fiber:** proprioception and motor function
 - **Small fiber:** pain and temperature
 - Motor fibers - letter
 - Sensory fibers - numerical
- Nerve fibers react to low oxygen (hypoxia) and local anesthetics
 - Anesthetic block C fibers (small, unmyelinated pain fibers)
 - Pressure affects A fibers (large fibers), so touch and motor control can be lost while pain remains
 - E.g., Saturday night paralysis - when you sleep on your arms and you are unable to move it but still feel the pain

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EXCITABLE TISSUE: MUSCLE

SKELETAL MUSCLE MORPHOLOGY

ORGANIZATION

- Built from individual muscle fibers (neurons to nervous system)

- Structure of skeletal muscles:

- Muscle fibers run parallel between tendons
 - Necessary in order for contraction

- Structure of muscle fiber

- Single, multinucleated
- Long and cylindrical shape
- Well-developed cross-striations and sarcolemma
- Requires nerve stimulation to contract
- No syncytial bridges between cells - don't need to work together
- Generally under voluntary control
- Fiber contains myofibrils - smaller structures inside
 - *Myofilaments* - smaller parts of myofibrils and contains proteins that allow the muscle to contract

Muscle -> muscle bundle -> fascicle -> muscle fiber -> myofibril -> actin and myosin filament

- Key protein for muscle contraction:

- Myosin - thick filaments; pulls on actin to create contraction
- Actin - thin filaments; slides past myosin
- Tropomyosin - covers actin's binding sites when muscle is relaxed
- Troponin - controls tropomyosin's position based on calcium levels
 - *Troponin T* - binds troponin complex to tropomyosin
 - *Troponin I* - inhibits actin-myosin interaction
 - *Troponin C* - binds to Ca to start contraction; bind to I and T

- Structural proteins keep the contractile proteins in the right position, and link the muscle structure to the extracellular matrix

- Z line - dark line in the middle of I band; anchors thin filaments
- A band - dark band with thick filaments (myosin) and some thin overlap
- H band - lighter area in the center of A band; thick filaments are present; region with no thin-thick overlap in relaxed muscle
- M line - dark line in the center of H band; where myosin polarity flips
- Sarcomere - unit of contraction, from one Z line to the next

- Filaments arrangement:

- Thick filaments (myosin II) - form A bands
- Thin filaments (actin, tropomyosin, troponin) - extend into I bands

- Myosin-II has:

- Two heads attach to actin
 - Actin-binding site
 - An ATPase (adenosine triphosphatase) site to break down ATP for energy

- Actin filaments are

- Double helix of actin chains
- Tropomyosin lies in the groove of its helix
- Troponin sits along tropomyosin and controls its access to actin

- Structural proteins:

- Actinin - binds actin to Z-line
- Titin (largest) - Connects Z-line to M-line which provides scaffolding and elasticity
 - Low resistance when unfolding, rapid resistance with further stretch protects sarcomere
- Desmin - Strengthens Z-line, then binds to plasma membrane

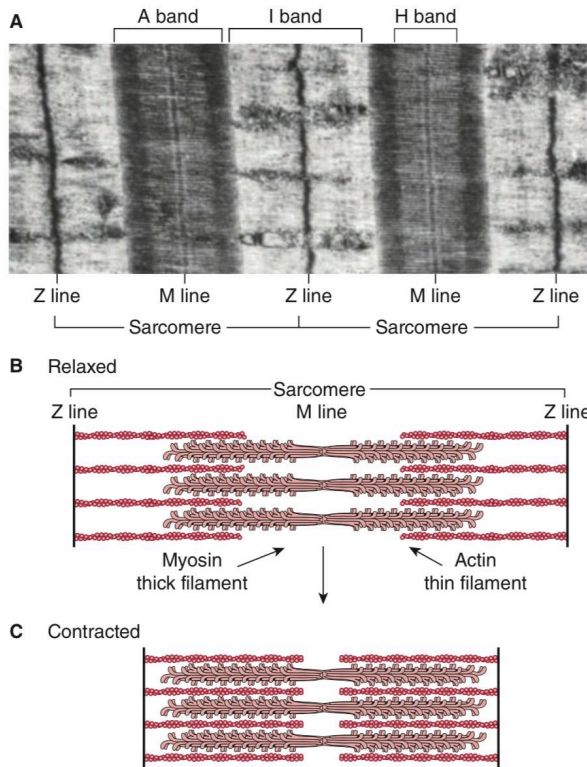
STRIATIONS

- Cross-striations - skeletal muscle's light and dark bands structure under microscope (light bands)

- Parts of a muscle sarcomere - basic contractile unit of skeletal muscle

- I band - light band with only thin filament (actin)

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SARCOTUBULAR SYSTEM

- Surrounds muscle fibrils; appears as vesicles & tubules in electron micrographs

- Two main parts:

- T system (transverse tubules)
 - Continuous with the sarcolemma and forms grid that weaves through the muscle fibrils
 - Space inside is an extension of extracellular space
 - Allows fast transmission of AP from cell membrane to all muscle fibrils, ensuring the whole fiber contracts at once
- Sarcoplasmic reticulum
 - Irregular curtain around each fibril
 - Terminal cisterns - enlarged areas formed between A and I band and is closely linked to T system
 - Triad - central T tubule lies between two SR cisterns; located at A-I junctions
 - Acts as a storage site for Ca ions and also helps in muscle metabolism

ELECTRICAL PHENOMENA & IONIC FLUXES

ELECTRICAL CHARACTERISTIC OF SKELETAL MUSCLE

- RMP of skeletal muscle is about **-90 mV**
- The absolute refractory period is 1-3 m/s long
 - After this, the muscle fiber experiences after-polarization, which are changes in membrane voltage that last longer and temporarily change how easy it is to excite the muscle again

NEUROMUSCULAR JUNCTION STRUCTURE

- **Motor neuron** - a nerve cell that stimulates muscle cells
- **Neuromuscular junction** - a synapse where a neuron connects with muscle fiber
- **Synapse** - refers to cell-to-cell junction between a nerve cell and either another nerve cell or an effector cell, such as in a muscle or a gland
- **Motor unit** - a group of muscle fibers that a single motor neuron stimulates
- **Presynaptic terminal** - end of a neuron cell axon fiber
- **Synaptic cleft** - space between presynaptic terminal and postsynaptic membrane
- **Postsynaptic membrane** - muscle fiber membrane (sarcolemma)
- **Synaptic vesicle** - a vesicle in the presynaptic terminal that stores and releases neurotransmitters
- **Neurotransmitters** - chemicals that stimulate or inhibit postsynaptic cells
- **Acetylcholine** - neurotransmitter that stimulates skeletal muscles

MOLECULAR BASIS OF CONTRACTION

- Sliding filament mechanism - muscle contraction happens where actin slide past myosin (filaments do not shrink, they overlap)
- **Sliding filament model**
 - A band stays the same length
 - Z lines move closer together during contraction, farther apart during relaxation
 - Contraction is powered by repeated cross-bridge cycles between myosin heads and actin
- **Cross-Bridge (Power Stroke) Cycling**

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- Energy for muscle contraction is supplied by ATP
- Energy is released by ATP → ADP and Pi, and energy from ATP is stored in myosin heads
- A new ATP must bind to myosin before cross-bridge is released
- **Rigor mortis** - occurs when a person dies and no ATP is available to release cross-bridges

CONTRACTION:

AP in axon terminal → opening of calcium voltage-gated → Ca entry → Ca binds to synaptic vesicle → neurotransmitter releases at synaptic cleft → ACh binds to sodium voltage-gated channels → AP in sarcolemma → AP in T tubules → activation dihydropyridine receptor which activates ryanodine receptor → opens calcium voltage-gated channel at SR → Ca enters cytoplasm → Troponin C gets calcium → Troponin C complex pulls tropomyosin, exposing actin binding sites → myosin heads attach to actin → hydrolysis of ATP to ADP → ADP pulls actin towards myosin (sliding filament mechanism; muscle contraction) → myosin heads detach → Troponin C detach calcium, covering the tropomyosin and actin binding site → Calcium ATPase or SERCA → pulls Ca back to the SR

CONTRACTILE RESPONSES

THE MUSCLE TWITCH

- Happens muscle fiber quickly contracted because a single action potential occurs in a muscle fiber, which is followed by relaxation

- Twitch begins about 2 ms after depolarization starts, even before the membrane finished repolarizing
- The length of the twitch depends on the type of muscle fiber:
 - Fast muscle fibers: lasts as short as 7.5 ms
 - Slow muscle fibers: last up to 100 ms

- Has 3 phases: latent phase, contraction phase, and relaxation phase

- Latent phase is the time between the application of a stimulus and the beginning of contraction

TYPES OF CONTRACTION

- Muscle do not always get shorter when contracting

- Isometric contraction (increase tension, same length)
 - The muscle generates force but does not change length
 - E.g., pushing against an immovable wall
 - Since work = force x distance, no work is done
- Isotonic contraction (increase tension, length shorten)
 - Muscle shortens while working against a constant load
 - E.g., lifting a dumbbell
 - Work is done because there is force and movement
 - Concentric - muscle tension increases as the muscle shortens
 - Eccentric - tension is maintained in a muscle, but opposing resistance causes the muscle to lengthen
- Negative work
 - Muscle lengthens while resisting a load
 - E.g., slowly lowering a weight

SUMMATION OF CONTRACTIONS

- Muscle nerves are electrically refractory
 - But contractile mechanism itself doesn't have a refractory period
- Summation - if the muscle receives another stimulus before it has fully relaxed, the new contraction adds on top of the first one (only muscle bundle)
- Recruitment - whole muscle working together
 - Total tension > single twitch
 - Increasing force of contraction by recruiting more muscle fibers
 - Slow fibers act first, then fast fibers
- Motor unit - consists of a single motor neuron and all the muscle fibers it innervates
 - AP in the neuron of a motor unit causes contraction of all the muscle fibers in that unit
 - Small, delicate muscles have very few fibers per motor unit
 - Large powerful, less precise muscles have fewer, large motor units
- Tetanus (tetanic contraction)
 - With rapid, repeated stimulation, the contraction fuse together into one smooth, sustained contraction
 - Incomplete tetanus - small relaxations still occur between stimuli

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- Complete tetanus - no relaxation at all; all contraction is fully fused -> tension is 4x greater than a single twitch
- Tetanus toxin
- Flaccid paralysis - no acetylcholine
- Role of stimulation frequency: If twitch duration = 10 ms
 - Stimulation frequency < 100 per second = separate twitches with full relaxation
 - Stimulation frequency > 100 per second = summation occurs, possibly leading to tetanus

MUSCLE TONE

- Constant tension produced by body muscles over long periods of time
- Muscle tone is responsible for keeping the back and legs straight, the head in an upright position, and the abdomen from bulging
- Muscle tone depends on a small percentage of all the motor units in a muscle being stimulated at any point in time, causing their muscle fibers to contract tetanically and out of phase with one another.

SKELETAL MUSCLE FIBER TYPES

- Fast-twitch fibers:
 - Contract quickly
 - Fatigue quickly
 - Use anaerobic respiration
 - Energy from glycogen
 - Light color
 - Used by sprinters
- Slow-twitch fibers:
 - Contract slowly
 - Fatigue slowly
 - Have a considerable amount of myoglobin
 - Use aerobic respiration
 - Dark color
 - Used by long distance runners
- Large postural muscles of the back and lower limbs contain more slow-twitch fibers

ENERGY FOR MUSCLE CONTRACTIONS

- Have three ATP dependent proteins:
 - Myosin head
 - Na/K ATPase to maintain RMP
 - Ca pump in the SR

- Muscle fibers store enough ATP to contract for about 5-6 seconds
 - If contraction is to continue beyond this time, more ATP must be produced
- ATP is derived from 4 processes in skeletal muscle
 - Conversion of two ADP to one ATP and one AMP by the enzyme adenylate kinase
 - Transfer of a phosphate from a molecule called creatine phosphate by the enzyme creatine kinase from ADP to form ATP
 - Anaerobic production of ATP during intensive short-term exercise
 - Aerobic production of ATP during most exercise and normal conditions
- Muscle fibers store enough ATP for 5-6 seconds of contraction
 - Next, ATP production by adenylate kinase and creatine kinase occurs. This is depleted after about 15 seconds
 - When a muscle fiber is working too strenuously for ATP stores and creatine phosphate to be able to provide enough ATP, anaerobic respiration predominates
 - Lactate produced by anaerobic fast-twitch fibers is used as a starting point for aerobic ATP production in slow-twitch fibers

MUSCLE FATIGUE

- Temporary state of reduced work capacity
- Without fatigue, muscle fibers would be worked to the point of structural damage to them and their supportive tissues
- Mechanism:
 - Acidosis and ATP depletion due to either an increased ATP consumption or a decreased ATP production
 - Oxidative stress, which is characterized by the buildup of excess reactive oxygen species (ROS; free radicals)
 - Local inflammatory reactions

MUSCLE SORENESS

- Followed by vigorous exercise, people sometimes experience muscle pain, which can last for several days
-

WEEK 1: INTEGRATIVE PHYSIOLOGY

OXYGEN DEFICIT

- Two distinct phases of O₂ use:
 - Oxygen deficit is the lag time between when a person begins to exercise and when they begin to breathe more heavily because of the exercise
 - Excess postexercise oxygen consumption is the lag time before breathing returns to its preexercise rate once exercise stops

ENERGY SOURCES & METABOLISM

PHOSPHORYLCREATINE (Creatine Phosphate)

- High-energy compound stored in muscle
- Carries high energy phosphate bond similar to ATP
- Breaks down as phosphate + creatine; creatine is used in body, and phosphate is attached to ADP to make ATP
- At rest:
 - Phosphorylcreatine store - some ATP in mitochondria transfers its phosphate to creatine
- During exercise:
 - Phosphorylcreatine is hydrolyzed to creatine and phosphate, releasing energy
 - Used to quickly form ATP from ADP, especially at the myosin-actin cross-bridges, so contraction can continue

CARDIAC MUSCLE MORPHOLOGY

- Cross-striated and functionally syncytial
 - Contracts rhythmically without external innervation due to pacemaker cells
 - Modulated by ANS
 - Numerous elongated mitochondria in close contact with muscle fibrils
 - Fibers branch and interdigitate
- Each fiber is a complete unit with its own cell membrane
- Z-lines present
 - Intercalated disks - Membranes of adjacent fibers form folds for strong cell-to-cell cohesion, which allows force from one cell to transmit to the next
 - Gap junction - found alongside intercalated disks which allows ions and excitations to spread quickly for contraction (syncytium function)

ELECTRICAL PROPERTIES

RESTING MEMBRANE AND ACTION POTENTIALS

- Changes in membrane permeability are responsible for producing APs and called pacemaker potential
- 1.) Phase 0 - rapid depolarization
 - Triggers opening of voltage-gated Na channels
 - Na rushes in -> membrane potential shoots upward
- 2.) Phase 1 - initial repolarization
 - Na channels becomes inactive
 - Certain K channels briefly open, cause K to efflux
 - Membrane potential dips slightly
- 3.) Phase 2 - plateau phase (**MOST IMPORTANT**)
 - Voltage-gated Ca channels open slowly and stay open; K channel close
 - Ca influx balances K efflux -> creates flat plateau
 - Critical for sustained contraction
- 4.] Phase 3 - rapid repolarization
 - Ca channels close
 - Multiple delayed-rectifier K channels open -> K efflux increases
 - Membrane potential returns toward -90 mV
- 5.) Phase 4 - resting potential
 - Stable RMP maintained by inward-rectifier K channels
 - Prepares cell for the next action potential

- K determines RMP

- Efflux during phase 1 and 3 controls repolarization
- Na had large influx during phase 0

MECHANICAL PROPERTIES

CONTRACTILE RESPONSE

- Contraction begins just after depolarization starts
- Excitation-contraction coupling
 - **SAME AS SKELETAL EXCITATION, BUT NO TROPONIN-TROPOMYOSIN COMPLEX INSTEAD CALCIUM-CALMODULIN COMPLEX LIKE IN SMOOTH MUSCLE**
- Refractory periods:
 - **Absolute** - phases 0-2 and first half of phase 3 cannot re-excite
 - **Relative** - second half of phase 3-4 can trigger another AP by strong stimulus but dangerous

WEEK 1: INTEGRATIVE PHYSIOLOGY

- Absolute and relative prevents tetanus (sustained contraction)
- Tetanization means the ventricles in the heart could not relax to fill with blood -> no circulation -> death
- The long AP and contraction overlap = built-in safety feature

CORRELATION BETWEEN MUSCLE FIBER LENGTH & TENSION

- Starling's Law of the Heart

- Initial fiber length = end-diastolic filling (preload)
- Developed ventricular pressure = end-diastolic volume
- As diastolic volume increases, so does the force of contraction -> up to an optimal point
- Tension decreases because of structural disruption of myocardial fibers when overstretched
- More blood in = stronger the contraction out

- Regulation of contractility - independent length

- Catecholamines (positive inotropy)
 - Epinephrine, norepinephrine
 - Increase contractility without changing muscle length
 - Makes sure heart contract and relax quicker, so there's more time for diastole

SMOOTH MUSCLE MORPHOLOGY

- No striations; no sarcomere
- Has actin and myosin II but no organized pattern
- **Single unit** - innervated by one neuron fiber
- Instead of Z lines, smooth muscle has dense bodies which is linked to alpha-actinin -> anchor actin filaments
- Tropomyosin is present but **tropoin is absent**
 - Doesn't use Ca regulation
- **SR is present but less extensive and developed**
- Fewer mitochondria so it relies more on glycolysis (anaerobic metabolism) instead of oxidative phosphorylation

TYPES

- Unitary (visceral) smooth muscle

- Found in large sheets (intestinal walls, uterus, ureters)
- Cells connected by gap junction -> low electrical resistance -> impulses spread easily
- Works like a syncytium (cells contract together as a single unit)
- Produces slow, wave-like contractions (good for moving contents in hollow organs)
- Fewer direct nerve endings
- Relies on gap junctions, hormones, and circulating substances - requires only 1 varicosity

- Multiunit smooth muscle

- Made of individual units
- Has direct nerve supply
- Produces fine, precise, and graded contractions
 - Iris, ciliary body, some large vessels
- Heavily dependent on autonomic innervation
- Doesn't need gap junction to contract; doesn't need to contract together unlike unitary
- Behaves more like skeletal muscle but still involuntary

- Mixed types

- Contains both unitary and multiunit smooth muscle

ELECTRICAL & MECHANICAL ACTIVITY

- Unitary smooth muscle

- RMP is unstable (-20 to -65 mV) - fluctuates with slow-waves and spikes (can overshoot 0 mV) sometimes
- Contract continuously and irregularly without nerve input
- Maintains partial contraction = muscle tone (tonus)
- Very slow excitation-contraction coupling
- **Have gap junctions that acts as a connection to act as one**

- Multiunit smooth muscle

- **Nonsyncytial** - cells act more independently
- No spontaneous rhythmic activity
- Contractions are more discrete, fine, stable, localized, and dependent on direct nerve input

- Involves K, Ca, Na, channels and Na/K ATPase

- Excitation-contraction coupling can have up to 500 ms delay (much slower than skeletal/cardiac muscle < 10 ms)

WEEK 1: INTEGRATIVE PHYSIOLOGY

ELECTRICAL ACTIVITY PATTERNS

- Slow sine wave-like fluctuations (few mV)
- Spikes (sometimes overshoot 0 mV, sometimes not)
- Spike duration is 50 ms; some have prolonged plateau like cardiac muscle

NEURONAL CONTROL

- Alpha-adrenergic receptor - contraction
- Beta-adrenergic receptor - relaxation

HUMORAL CONTROL

- Angiotensin II
- ADH (vasopressin)
- Epinephrine
- ANP

MYOGENIC AUTOREGULATION

- Primarily found in blood vessels, specifically at afferent glomerular arterioles

LOCAL HUMERAL CONTROL

- Autocrine - hormone act on itself or its part
- Paracrine - hormone act on different parts
- Vasodilator - adenosine

MOLECULAR BASIS OF CONTRACTION

- Still needs Ca
- No troponin C so Ca binds to calmodulin - forms Ca-calmodulin complex
- Myosin is dephosphorylated by myosin light chain phosphatase (MLCP)
- Because of latch-bridge mechanism, myosin remains bound to actin even after Ca falls
 - Allows sustained contraction with low energy use - important for blood vessels
- Neurotransmitters effect (unitary):
 - Norepinephrine
 - Hyperpolarizes membrane
 - Decrease spike frequency
 - Muscle relaxes
 - Mediated by adrenergic receptors -> inhibitory effect in intestine
 - Acetylcholine
 - Depolarizes membrane
 - Increase spike frequency

- Muscle becomes more active (increase tone and rhythm contraction)
- Works via phospholipase C
- Neurotransmitters effect (multiunit)
 - Also controlled by ACh and NE, but NE persists longer -> irregular tetanus
 - If twitches, it's like skeletal muscle twitch but 10x longer

SAME CONTRACTILITY MECHANISM AS SKELETAL, BUT T TUBULE IS REPLACED BY CAVEOLAE, AND TROPONIN-TROPOMYOSIN COMPLEX IS REPLACED BY CALCIUM-CALMODULIN COMPLEX WHICH ACTIVATES MYOSIN FOR HYDROLYSIS, THEN SO ON, AND SO FORTH

MORE CALCIUM INSIDE CYTOPLASM FOR CONTRACTION

NOREPINEPHRINE AND EPINEPHRINE ARE THE NEUROTRANSMITTERS PRESENT

WEEK 1: INTEGRATIVE PHYSIOLOGY

FROM RECENT PPT

- Occurs to do any form of voluntary and involuntary activity

- Normal cell: -70
- Large nerve cells: -90

- EP must exist across membranes of cells for the signals to be sent to any cell

- Once electrical potential is achieved, electrochemical impulses are generated which carry signals to the cell

- Concentration of extracellular and intracellular membrane of the cell is different

- Na - 40-60
- Cl - -70mV
- K - -90mV

- RMP - electrochemical state of cell at rest

- -90mV
- Depends on concentration gradient and permeability of the ion across the membrane
- Equalize concentration: K moves out of the cell. It diffuses intracellularly through the K leak channel
- Basis for cell-to-cell communication
- Keeps cell "in ready state" to fire an AP

- Na/K pump - main workforce responsible for establishing concentration gradients of K and Na

- Goldman equation - calculate diffusion potential when the membrane is permeable to several different ions

- Nernst of K = 94 mV; Na = 61mV - EP needed in order to attract the ions back to their corresponding places
- the pump will contribute -4 mV which will result in a net RMP of -90 mV

- AP

- -70 at rest
- -55 threshold - Na voltage gated slowly opens
- 35 depolarizes - Na voltage gated are open, K slowly open (depolarization)
- Repolarization - K are open, Na starts to close

- Hyperpolarization - All Na voltage are closed, K still open; AP is inhibited (Absolute refractory period)
- Back to RMP

- Na voltage gated channel

- Activation gate - near outside
- Inactivation gate - near the inside
 - Higher K concentration

- K voltage gated channel

- Resting stage - membrane potential: -90 mV

- Resting stage

- Polarized: -70mV

- Propagation

- AP travel along the nerve
- Moves in a forward direction
- Myelin sheath - prevents lost of ion

- All-or-none

- Nerve is not dependent on the strength of the stimulus

- Saltatory Conduction

- 5 to 50 fold speed
- Large nerves are usually myelinated

Skeletal Muscle

- Composed of myofibrils

- 10-80 um in diameter

- innervated by only one nerve ending